제 1 장

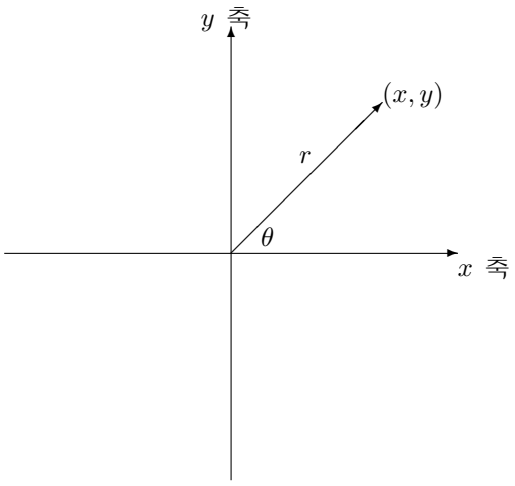
좌표계

1.1 2 차원과 3 차원에서의 직교 좌표계

1. 2 차원 직교좌표계의 면적소 (area element in the 2 dimensional Cartesian coordinate system) : $da = dxdy$.
2. 3 차원 직교좌표계의 부피소 (volume element in the 3 dimensional Cartesian coordinate system) : $d\tau = dxdydz$.

1.2 평면 극좌표계

1. 평면 극좌표계 (plane polar coordinate system in 2 dimension) :



$$\sin\theta = \frac{y}{r}, \qquad \cos\theta = \frac{x}{r}. \tag{1.1}$$

$$x = r\cos\theta, \qquad y = r\sin\theta. \tag{1.2}$$

$$\begin{aligned} \theta &= \tan^{-1}\left(\frac{y}{x}\right) = \arctan\left(\frac{y}{x}\right) = \mathrm{atan}\left(\frac{y}{x}\right), & (1.3)\\ r &= \sqrt{x^2+y^2}. & (1.4) \end{aligned}$$

$\sin^2\theta = (\sin\theta)^2$ 이지만 $\sin^{-1}\theta \neq (\sin\theta)^{-1}$ 임을 주의하라.

2. 평면 극좌표계의 면적소 (area element) : $da = dxdy = rdrd\theta$

3.

$$I = \int_{-\infty}^{\infty} dx e^{-x^2} = \int_{-\infty}^{\infty} dy e^{-y2}. \tag{1.5}$$

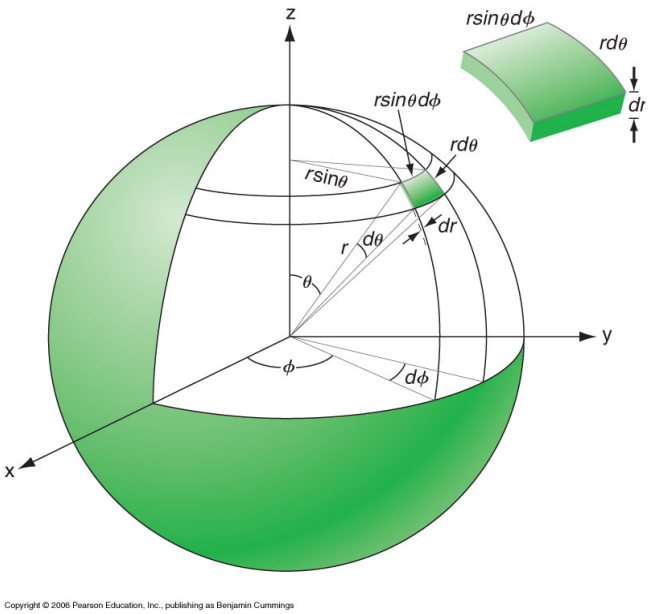
$$\begin{aligned} I^2 &= \int_{-\infty}^{\infty} dx x^{-2} \times \int_{-\infty}^{\infty} dy y^{-2},\\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dxdy e^{-x^2-y^2},\\ &= \int_0^{2\pi}\int_0^{\infty} rdrd\theta e^{-r^2},\\ &= \int_0^{2\pi} d\theta \int_0^{\infty} drre^{-r^2},\\ &= 2\pi\frac{1}{2} = \pi. \end{aligned} \tag{1.6}$$

$$\therefore I = \sqrt{\pi}. \tag{1.7}$$

1.3 구면 극좌표계

1. 구면 극좌표계 (spherical polar coordinate system in 3 dimension) :

$$\begin{aligned} x &= r\sin\theta\cos\phi, & (1.8)\\ y &= r\sin\theta\sin\phi, & (1.9)\\ z &= r\cos\theta. & (1.10) \end{aligned}$$



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2.

$$\frac{y}{x} = \frac{r\sin\theta\sin\phi}{r\sin\theta\cos\phi} = \tan\phi \Rightarrow \phi = \tan^{-1}\left(\frac{y}{x}\right) = \arctan\left(\frac{y}{x}\right) = \operatorname{atan}\left(\frac{y}{x}\right), \tag{1.11}$$

$$r = (x^2+y^2+z^2)^{1/2}, \tag{1.12}$$

$$\cos\theta = \frac{z}{r} \Rightarrow \theta = \cos^{-1}\frac{z}{(x^2+y^2+z^2)^{1/2}}. \tag{1.13}$$

3. 부피소 (volume element) : $d\tau = (dr)(rd\theta)(r\sin\theta d\phi) = r^2\sin\theta dr d\theta d\phi$

4. ∇^2 : Laplacian 기호 ("del squared" 라고 발음)

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}, \tag{1.14}$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}. \tag{1.15}$$

1.4 복소수

1. 복소수 : $z = x + iy$

2. 복소수 평면에서 직교좌표의 극좌표로 변환 :

   평면 극좌표계에서

$$x = r\cos\theta, \qquad y = r\sin\theta. \tag{1.16}$$

이므로

$$z = x + iy = r\cos\theta + ir\sin\theta = r(\cos\theta + i\sin\theta) = re^{i\theta}. \tag{1.17}$$

위 식에서

$$r = \sqrt{x^2 + y^2},$$
$$\tan\theta = \frac{x}{y} \Rightarrow \theta = \tan^{-1}\frac{x}{y}.$$

3. 켤레복소수 (complex conjugate)

$$\begin{aligned} z^* &= (x+iy)^* = x - iy = r\cos\theta - ir\sin\theta, \\ &= r(\cos(-\theta) + i\sin(-\theta)) = re^{-i\theta}, \qquad (1.18) \\ |z|^2 &= zz^* = (x+iy)(x-iy) = x^2 + y^2 = r^2 \Rightarrow r = |z|. \qquad (1.19) \end{aligned}$$

$$e^{ix} = \cos x + i\sin x. \tag{1.20}$$

4. Euler 공식 (Euler's formula)

Maclaurin 급수 (Maclaurin series) :

$$\begin{aligned} f(x) &= a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n + \cdots, \\ f(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^n(0)}{n!}x^n + \cdots \qquad (1.21) \end{aligned}$$

♣ **예제 1** :

$$\begin{aligned} \cos\theta &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots, \qquad (1.22) \\ \sin\theta &= \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots. \qquad (1.23) \end{aligned}$$

♣ **예제 2** :

$$\begin{aligned} e^{i\theta} &= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{\theta^5}{5!} + \cdots, \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots\right). \qquad (1.24) \end{aligned}$$

5. 위의 (1.22), (1.23) 식들과 (1.24) 식을 비교하면 다음의 **오일러 공식**을 얻을 수 있다.

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta.} \tag{1.25}$$

6. De Moivre 공식 (De Moivre's formula) :

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta). \tag{1.26}$$

♣ **예제 1** Euler 공식과 De Moivre 공식을 이용하여 $(1-i)^8$ 의 실수값과 허수값을 구하여라.

♣ **예제 2** Euler 공식을 이용하여 다음 적분을 증명하여라.

$$\int_0^\infty dx e^{-ax}\cos bx = \frac{a}{a^2+b^2} \tag{1.27}$$

$$\int_0^\infty dx e^{-ax}\sin bx = \frac{b}{a^2+b^2}. \tag{1.28}$$